

## 7.8 ROTOR SYSTEM

### 7.8.1 Main Rotor (see Fig. 7-11)

The main rotor ("System Bölkow") comprises a rigid-mounted head with four glassfiber reinforced plastic (GRP) blades.

### 7.8.2 Main Rotor Head (see Fig. 7-11)

The rigid main rotor head is bolted directly to the main rotor mast and consists of a star hub assembly and four blade-mounting forks with pitch horns. The hub assembly consists of a single-piece cross-shaped drop forged titanium hub housing the blade retaining system with tension-torsion straps and mounting the oil reservoir and feathering hinge bearings. Each blade-mounting fork with inner sleeve assembly is inserted into an arm of the hub and retained by a tension-torsion strap. The blade-mounting forks permit blade pitch changes in response to flight control movements.

### 7.8.3 Main Rotor Blades (see Fig. 7-11)

Each main rotor blade is a constant-cord asymmetrical airfoil. The blades are constructed of a diagonally-layered fiberglass cloth outer skin covering a C-spar made of a unidirectional fiberglass roving lamination which forms a loop around the blade root, and a core of hard foam filler. The blade leading edge is protected against erosion and foreign object damage by titanium strips. A blade fitting assembly is bolted over the blade root and provides mounting of the blade to the head. Two trim tabs are provided on the trailing edge of the outer portion of each blade to allow for tracking adjustment.

Aerodynamic motions of lead/lag and flapping are absorbed by the blade inherent elasticity. Each blade may be fitted with pendulum absorbers near the root to reduce inflight vibrations.

### 7.8.4 Tail Rotor Assembly (see Fig. 7-12)

The tail rotor is of the semi-rigid teetering design. The two-bladed rotor fastened in a delta hinged hub provides antitorque action and directional control. A titanium shaft having a fork head and splined end houses two spherical teetering bearings and drives the rotating part of the yaw control system sliding sleeve. The tail rotor shaft is splined to and driven by the tail rotor output bevel gear.

Blade centrifugal forces are carried by laminated steel tension straps that attach outboard to the blade mounting forks and inboard to a strap shoe which is held centered by the tail rotor assembly mounting bolt. The blade mounting forks rotate about the hub on Teflon bushings. Dynamic weights are installed on the control yoke to reduce control pedal forces. These weights are adjusted to compensate for main rotor torque required at a hover.

The tail rotor blades are constructed similar to the main rotor blades. The blades are retained in the mounting forks by two bolts.

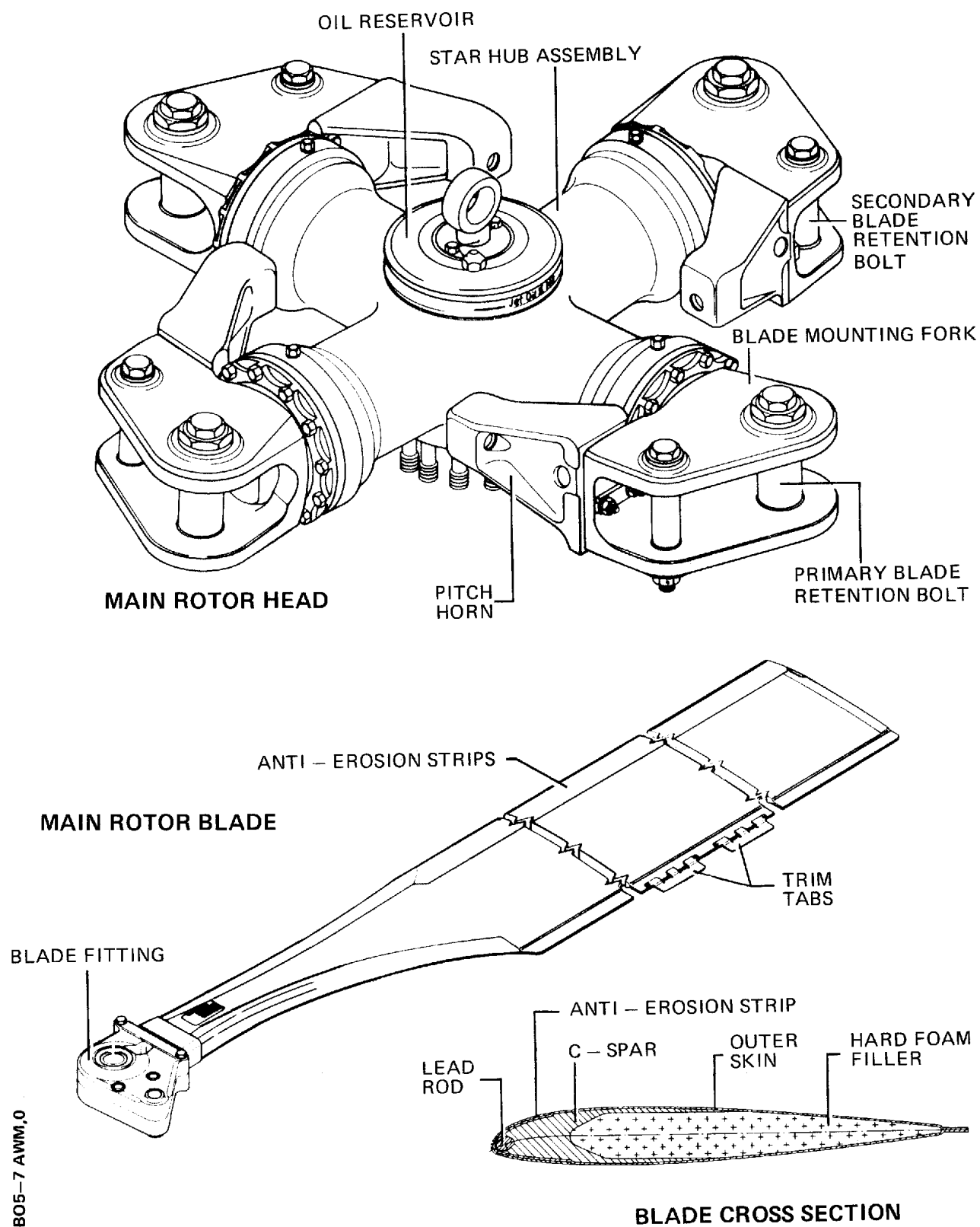
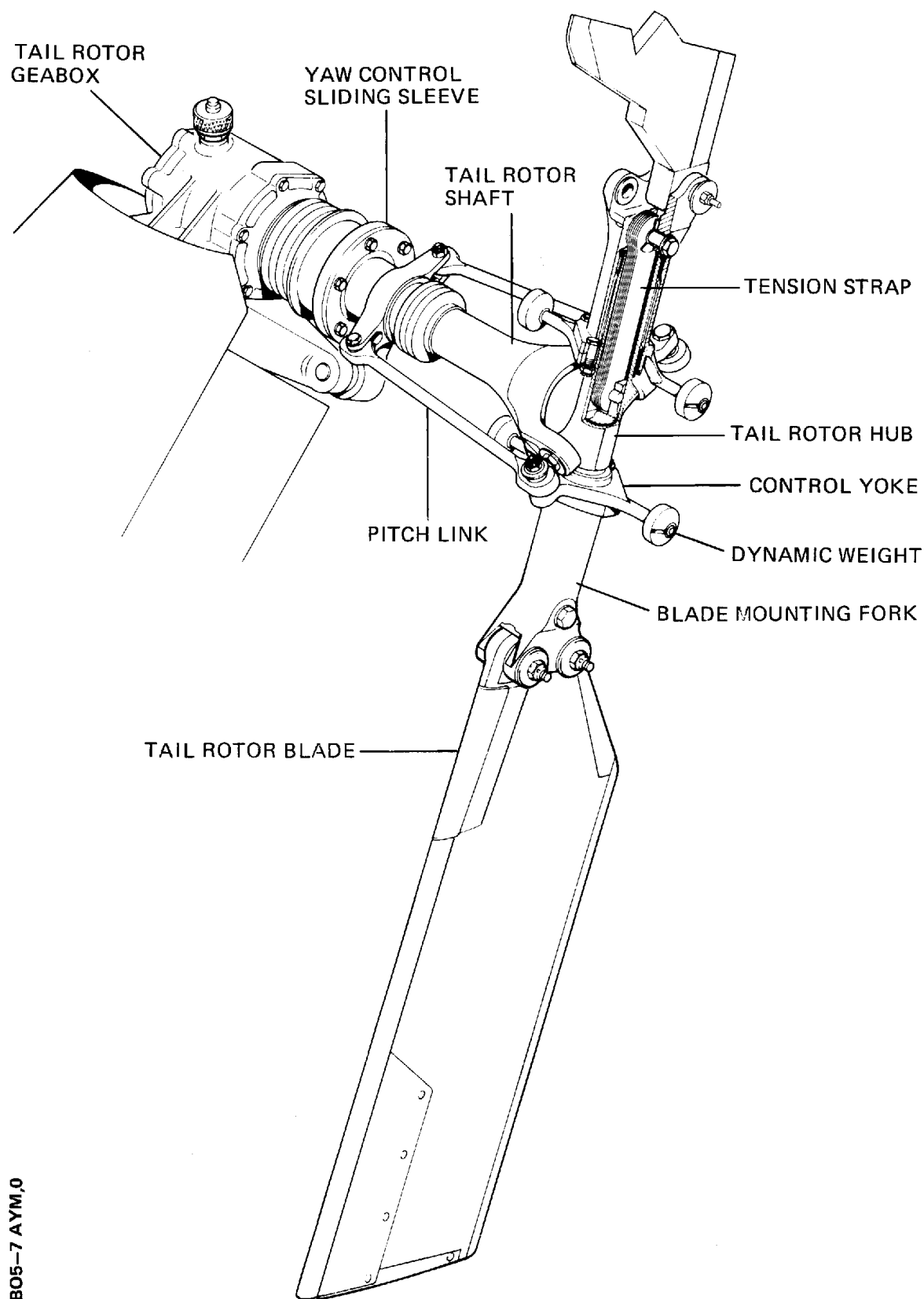


Fig. 7-11 Main Rotor



BO5-7 AYM,0

Fig. 7-12 Typical Tail Rotor Assembly